

Field Trial of Commercial 1295-nm O-Band DV-QKD under Coexistence with Classical Channels in a High-Capacity ROADM-based Metropolitan Production Fiber Network

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Abstract We report field deployment of a 1295.56 nm O-band DV-QKD system coexisting with high-capacity C-band traffic in a metropolitan link. Measurements under varying channel allocations and powers show robust coexistence, improved tolerance over 1310 nm operation, and no degradation in classical C-band performance or capacity. ©2026 The Author(s).

Introduction

Quantum key distribution (QKD) [1] is increasingly regarded as a promising technology for securing optical communications in future network infrastructures. However, its large-scale adoption still relies on the ability to operate over shared fiber infrastructure without requiring dedicated (dark) fibers, which makes coexistence of weak quantum signals with high-power classical traffic an engineering challenge [2].

Among the different integration strategies, dense wavelength-division multiplexing (DWDM) is broadly adopted in the architecture of current metropolitan transport networks [3,4]. In such scenarios, spontaneous Raman scattering (SpRS) generated by classical channels (CC) is one of the main impairments limiting the performance of QKD systems [5]. While CV-QKD can be integrated within the C-band in such DWDM systems, O-band discrete-variable (DV) QKD is also an attractive solution, benefitting both from the large separation between the quantum channel (QC) and CCs and from operating on the anti-Stokes side of the Raman spectrum. Consequently, it is very well suited for coexistence in conventional high-power C-band communications systems [6], while keeping the C-band fully available for classical traffic without reserving channels or introducing guard bands [7,8]. This makes coexistence more practical in production networks, at the expense of higher O-band fiber attenuation, which is not limiting for the link lengths typically found in metro networks.

Although most O-band QKD implementations operate at 1310 nm, the wavelength of choice within the O-band remains an important design parameter. Shifting the QC toward shorter O-band wavelengths can further reduce the Raman noise reaching the receiver while preserving compatibility with DWDM components [9].

In this paper, we report the deployment and evaluation of a 1295.56 nm time-bin BB84 DV-QKD system in a metropolitan production network, assessing its coexistence with high-capacity C-band traffic by varying CC number, spectral allocation, and launch power. We also benchmark it against a conventional 1310 nm implementation to quantify the improved coexistence margin in a field environment.

Experimental setup

The coexistence tests are carried out over several links of the TEFQCI network, Telefónica's Quantum Communications research network deployed on production fibers in the Madrid region, which integrates into the larger-scale MadQCI network [4], a heterogeneous, multi-tenant, multi-technology network, installed in production facilities and sharing the infrastructure with commercial traffic [3]. Recently, the Distrito-Norte-Concepción links in TEFQCI were migrated from a fixed optical add/drop multiplexer (FOADM) based ring to a carrier-grade extended C-band colorless, directionless reconfigurable optical add/drop multiplexer (ROADM) based horseshoe topology, with both end nodes located at the Telefónica CDO/CTIO lab in Distrito. This change required additional link amplification, detrimental to the coexistence of QC and CCs. The fiber layout for the TEFQCI ROADM-based horseshoe topology is depicted in Fig. 1(a). The two end nodes at Distrito are connected to Norte (15.0 km, 8.3 dB) and Concepción (15.3 km, 9.2/14.3 dB) via standard ITU-T G.652-D single-mode fiber pairs, while Norte and Concepción are linked by an 8.1 km fiber pair (2.9/2.4 dB). As represented in Fig. 1(b)—only showing the physical layer infrastructure—, all links support QKD injection in the C and O bands using off-the-

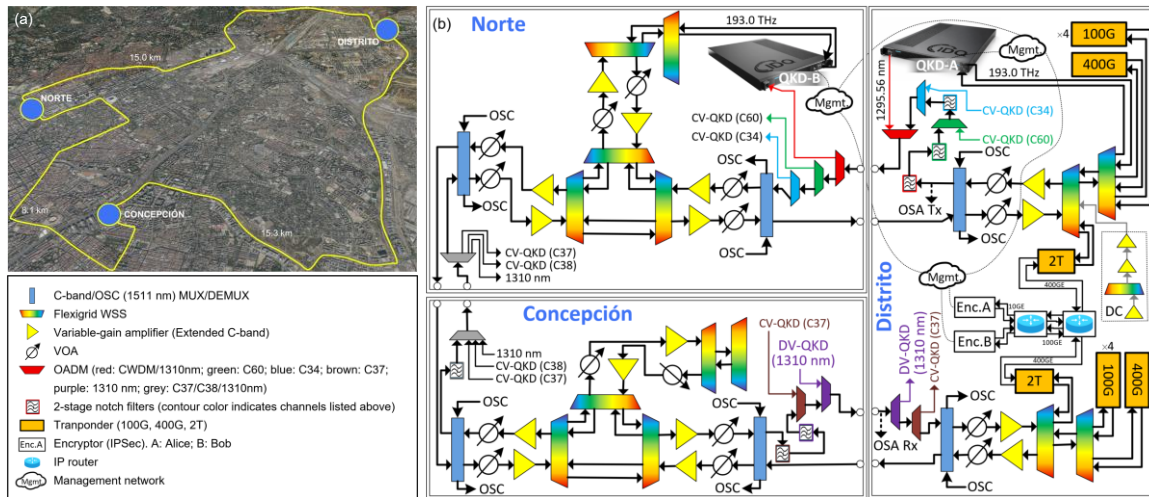


Fig. 1: Experimental setup. (a) Field-deployed fiber layout of the Telefónica QKD research ring TEFQCI. (b) Schematic of the ROADM-based arrangement used to test the coexistence of CCs with IDQ's Clavis XG QKD system operating at 1295.56 nm.

shelf optical add/drop multiplexers (OADMs) and notch filters (see legend in Fig. 1), which remove ASE noise within the slot reserved for the QC prior to multiplexing, with attenuation averaging out at ~ 0.6 dB per filtering stage. The ROADM network is managed via the optical supervisory channel (OSC), at 1511 nm, while servers, switches, and quantum equipment are managed over a 2.5G channel at 193.9 THz (Distrito–Norte) and 193.1 THz (Distrito–Concepción).

The QKD system under test is an ID Quantique Clavis XG operating in the O-band. This DV-QKD system implements the time-bin BB84 protocol with two-decoy states and composable finite-key analysis [10], including full protocol execution, key sifting, error correction, error verification and privacy amplification, and is designed for DWDM environments and coexistence with CCs. A distinctive feature is the use of a 1295.56 nm QC instead of the more conventional 1310 nm wavelength. It is deployed in the Distrito–Norte link through a standard CWDM OADM with a 1310nm port bandwidth of 100 nm (1260–1360 nm) and >55 dB isolation.

For the coexistence experiments, CCs are injected into the node at Distrito facing Norte and received at Distrito via the node connected to Concepción. In general, we consider unidirectional transmission of emulated 400G channels generated via a dummy channel (DC) generator, implemented with a wavelength-selective switch (WSS) and several erbium-doped fiber amplifiers (EDFA), in power-lock and gain-lock modes. The considered spectral band spans from 191.3 THz (1567.13 nm) to 196 THz (1529.55 nm), and is populated with up to 44 CCs, including up to 37 DCs. Among the active channels, there are four 100G, one 400G channel, and one 2T channel, as well as Clavis XG 10G service channel (SC) at 193 THz, providing synchronization, optical and temporal

alignment, key distillation, and post-processing, and the aforementioned 2.5G network management channels (C31, C39).

Fig. 1(b) also depicts the encryptors (Enc-A, Enc-B), which connect to QKD-A and QKD-B over an mTLS session via the management network to retrieve keys from the QKD modules through the standard ETSI GS QKD-014 interface [11]. Keys are then used for encrypted data transmission between the two end nodes at Distrito. Due to the QKD modules and encryptors being located at different sites, this key retrieval mechanism does not align with standard security procedures and was implemented solely for experimental validation and to test physical-layer capabilities of the equipment under test.

Results

Fig. 2 shows the DV-QKD performance when the 1295.56 nm QC coexists with varying numbers of 400G, 100G, and DCs in the extended C-band.

In Fig. 2(a), the total launch power is kept constant at ~ 16.5 dBm and distributed among an increasing number of channels (from 2 to 42, excluding the SC and management channels mentioned above). The goal of this test is to evaluate the impact of distributing power across the entire band or concentrating it in certain spectral regions. Note that the maximum power output of the EDFA at Distrito toward Norte is 22.5 dBm, reduced by the VOA and OSC MUX, as well as a number of filtering stages for QC injection, as shown in Fig. 1(b), the latter introducing a total loss of 4.8 dB. We observe that concentrating 16.5 dBm in a small number of CCs at higher frequencies (orange) results in the worst system performance (5.6 kb/s SKR, 2.4% QBER), as a result of the reduced spectral separation between these high power CCs and the O-band. In contrast, concentrating the power at lower frequencies (blue) provides the best

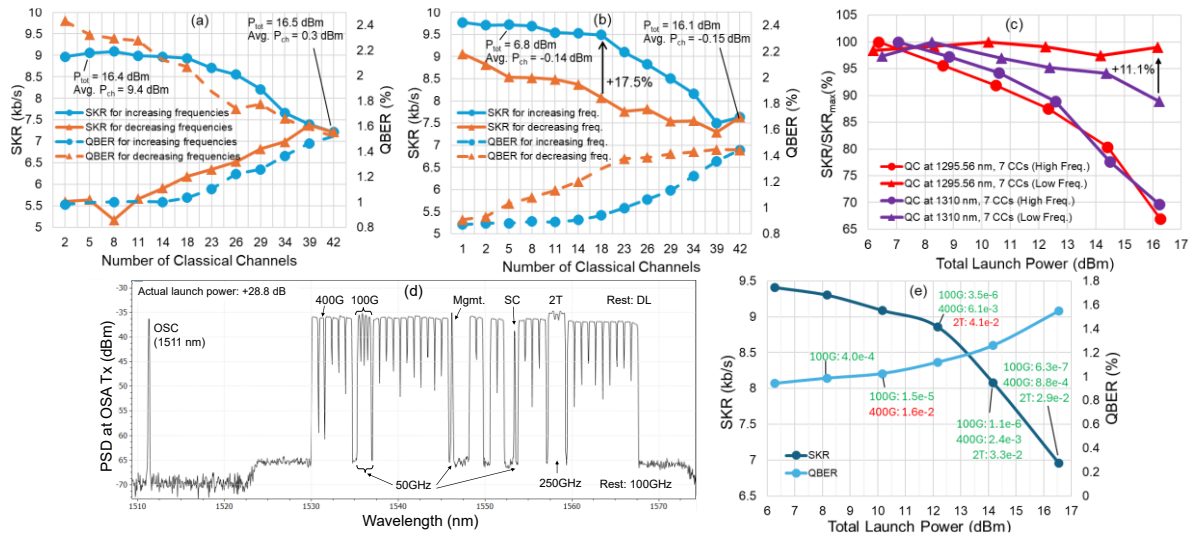


Fig. 2: (a)–(b) QKD performance in terms of SKR (solid) and QBER (dashed) for a QC co-propagating with an increasing number of CCs, filled from lower to higher frequencies and vice versa. In (a), the total power is kept constant at ~16.5 dBm. In (b), the per-channel power is kept constant at ~0 dBm. (c) SKR reduction as the power of localized channels at higher or lower frequencies increases, for the 1295.56 nm system (Distrito–Norte) and a 1310 nm system (Concepción–Distrito). (d) Channel plan and power spectral density (PSD) at the output of Distrito toward Norte, before QC multiplexing. (e) QKD performance for the coexistence scenario in (d), along with pre-FEC BER for 100G, 400G, and 2T channels (red indicates service down).

performance (~9 kb/s SKR, 0.98% QBER). Note the negative vs. positive SKR slope for the two power distribution directions.

In contrast, in Fig. 2(b) the per-channel power is kept constant (~0 dBm), representing a practical scenario from a telecom operator perspective, where channels are transmitted with a fixed carrier-grade per-channel power. As above, the C-band can be filled starting from low to high frequencies (blue) or from high to low frequencies (orange). Results show that filling the C-band from low to high frequencies provides the best performance, with a maximum improvement of 17.5% when 18 channels are transmitted.

To evaluate the impact of shifting the QC from 1310 nm to 1295.56 nm, Fig. 2(c) compares the normalized SKR of the 1295.56-nm Clavis XG system (Distrito–Norte) with a 1310 nm DV-QKD system (Concepción–Distrito). Both links exhibit comparable attenuation, and the launch powers were equalised to ensure a fair comparison.

The QC coexists with seven DCs placed at either the low- or high-frequency end of the extended C-band, while the per-channel power is varied. For high-frequency CCs, both systems show a ~30% SKR reduction as power increases from >6 to ~16.5 dBm. For low-frequency CCs, the 1295.56 nm system remains largely stable, whereas the 1310 nm one degrades by ~11.1%.

Finally, in Fig. 2(d)–(e) we show the OSC and C-band CC distribution measured at the OSA Tx monitoring point in Fig. 1(b), as well as the QKD performance and pre-FEC BER for a selected 100G, 400G and 2T channels, as the per-channel power is varied under full C-band loading. The 100G, 400G, and 2T channels use QPSK (50

GHz spectral width), 16QAM (100 GHz), and 64QAM (250 GHz), respectively. We observe a reduction in SKR of ~25% when the total power goes down from 16.5 to 6.5 dBm. However, the 2T channel can only support a power drop of 3–4 dB (equivalent to 15% SKR reduction), while the 400G channel can support a power reduction of up to 5–6 dB (7% SKR reduction). Further power reduction leaves only 100G channels operational, but the marginal SKR gain does not justify the resulting CC performance degradation.

Using the highest total power in Fig. 2(e), providing optimal CC performance, we conclude the study with the successful transmission of encrypted 10GE traffic over a 400GE signal carried on the 2T channel between the two end nodes at Distrito. Encryption is implemented via an IPsec (L3) tunnel. To traverse the 400GE link, the encryptors are configured through a routing policy that abstracts network complexity while ensuring end-to-end encryption.

Conclusions

In this work, we have deployed and demonstrated the coexistence limits of an O-band (1295.56 nm) QKD system with different classical channel configurations and launch powers in a production metro network environment. The results show that a shorter-wavelength O-band operation improves tolerance to Raman noise and enables a more robust coexistence than a conventional 1310 nm implementation when long-wavelength C-band channels are transmitting. This supports 1295.56 nm DV-QKD as a practical solution to integrate quantum security into metro networks without sacrificing C-band capacity.

Acknowledgments

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